

HVAC parts list - 2168 Blind Bay Rd, Shaw Island WA 98286 (San Juan County) - duplex

BUILD-24-0020 model 2168-blindbay-hvac printed 2026-09-25 ductless heat 24,719 Btuh cooling 1.23 t 2 outdoor 7 heads 0 ducted units 0 registers line sets 285 ft duct 0 ft vent 44 ft 11 days on site
Design 26 F / 77 F (shaw island: STAND-IN: MJ8 Table 1A Friday Harbor AP 30 F / 75 F [V] shifted by Bellingham's RC-1 gap (-4 / +2 F) - pull Table RC-1 (Shaw is 3 mi from the Friday Harbor station)). Foundation crawl, line sets concealed, ducts inside, ventilation exhaust. Quantities only - the prices are set on the price list in the app.

Loads

heating 24,719 Btuh (12.4 Btuh/sf) · cooling 13,051 + 1,759 latent · WSU sheet 25,508 Btuh · infiltration 5 ACH50 -> 0.339 ACH at design · ventilation 112.9 cfm (exhaust)

room	storey	sf	heat Btuh	cool Btuh	glass sf	ext wall ft
BED 2	main	121	2,123	1,892	77	27
LIVING A	main	191	1,975	881	47	28
PRIMARY BED	main	119	1,777	1,152	39	29
BED 1	main	125	1,751	1,772	53	24
BED CLOSET	main	54	1,470	253	0	23
LIVING B	main	150	1,239	151	0	24
KITCHEN	main	99	1,209	1,549	11	17
ENTRY	main	49	1,208	948	30	14
DINING	main	172	1,177	650	33	12
KITCHEN	main	123	945	1,763	30	9
HALL	main	42	792	78	0	16
BED 1	main	122	770	351	0	13
BED 2	main	124	769	353	0	13
STORAGE	main	24	751	577	21	13
BED CLOSET	main	52	709	54	0	16
ENTRY	main	104	633	103	0	11
DINING	main	103	495	101	0	7
BATH	main	52	354	52	0	6
BATH	main	52	339	51	0	6
CLOSET	main	21	211	21	0	4
PR BATH	main	21	208	21	0	4
STORAGE	main	33	169	33	0	2
PRIMARY CLOSET	main	47	105	45	0	0

The system

unit A - ductless: heat 12,106 Btuh, capacity at design 30,117 Btuh

outdoor eq_mz_30 30k (multi) on the W wall, at the electrician's Ductless mini-split outdoor unit (unit B) (LIVING A, electrical/jobs/shaw/devices.json); 165 ft of line set

head in	zone rooms	carried	heat	cool
LIVING A	ENTRY, LIVING A, DINING, KITCHEN		4,913	2,848
BED 1	BED 1	STORAGE	3,037	2,349
PRIMARY BED	PRIMARY BED	PR BATH, PRIMARY CLOSET	2,537	1,218
BED 2	BED 2	CLOSET, BATH	1,619	426
head	room	storey	heat	cool
6k	LIVING A	main	4,913	2,848
6k	BED 1	main	3,037	2,349
6k	PRIMARY BED	main	2,537	1,218
6k	BED 2	main	1,619	426

unit B - ductless: heat 12,613 Btuh, capacity at design 24,094 Btuh

outdoor eq_mz_24 24k (multi) on the E wall, at the electrician's Ductless mini-split outdoor unit (unit A) (LIVING B, electrical/jobs/shaw/devices.json); 120 ft of line set

head in	zone rooms	carried	heat	cool
DINING	LIVING B, KITCHEN, DINING	ENTRY, STORAGE, HALL	6,522	3,409
BED 2	BED 2	BED CLOSET, BED CLOSET, BATH	5,224	2,250
BED 1	BED 1		867	351

head	room	storey	heat	cool
9k	DINING	main	6,522	3,409
6k	BED 2	main	5,224	2,250
6k	BED 1	main	867	351

Runs (measured, before waste)

system / size	LF
condensate 0.75	74
dryer 4	20
exhaust 4	8
exhaust 5	4
hood 6	11
refrigerant 0.25x0.375	285

Equipment

Equipment

item	measured	order	unit	note
Ductless wall head 6k	6	6	EA	LIVING A: zone ENTRY, LIVING A, DINING, KITCHEN - heat 4,913 / cool 2,848 Btuh · BED 1: zone BED 1 - heat 3,03
Ductless wall head 9k	1	1	EA	DINING: zone LIVING B, KITCHEN, DINING - heat 6,522 / cool 3,409 Btuh
Multi-zone outdoor unit - 24k, cold-climate, up to 3 indoor units	1	1	EA	multi-zone outdoor unit, 24k nominal, ~24,094 Btuh at 26 F (ResStock cold-climate curve), at the electrician's
Multi-zone outdoor unit - 30k, cold-climate, up to 4 indoor units	1	1	EA	multi-zone outdoor unit, 30k nominal, ~30,117 Btuh at 26 F (ResStock cold-climate curve), at the electrician's
Wired wall controller per ductless zone (optional) (option)	7	7	EA	wired wall controller per head (option)

Rough-in

Condensate

item	measured	order	unit	note
Condensate line 3/4 in PVC (or 5/8 vinyl with the line set)	74.3	82	LF	condensate: gravity, 1/8 in per ft, to daylight (M1411.3) - never into the crawl

Refrigerant piping

item	measured	order	unit	note
Anti-vibration pads (set of 4)	2	2	EA	
Communication cable 14/4 stranded (outdoor unit to each indoor unit)	285	314	LF	with the line set (powers and talks to the head)
Equipment pad - composite, sized to the outdoor unit	2	2	EA	outdoor unit pad
Line set 1/4 x 3/8 insulated (pre-flared or brazed)	285	314	LF	LIVING A 6k head, 13 ft concealed · BED 1 6k head, 71 ft concealed · PRIMARY BED 6k head, 34 ft concealed · BE
Outdoor unit stand / wall bracket (snow and drainage clearance)	2	2	EA	snow / drainage stand (NW practice: 12-18 in clear of grade)
Wall penetration - 3 in sleeve, sealed and flashed	2	2	EA	rim-joist penetration for the line-set bundle beside the outdoor unit

Support + protection

item	measured	order	unit	note
Bore through a stud / plate (counted, not priced)		14	14	EA plates the line set passes (bottom plate + subfloor per storey)
Fire caulk / sealant (tube)		2	2	EA fire caulk / sealant at the penetrations
Head / controller blocking (framer or HVAC)		7	7	EA head mounting plate blocking
Nail plate at a line-set / cable bore within 1-1/4 in of the face		14	14	EA nail plates at those bores

Ventilation and exhaust

Ventilation

item	measured	order	unit	note
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Bath fan exhaust duct 4 in insulated flex (50 cfm fan: <= 25 ft, Table M1505.4.4.2)	8.5	10	LF	PR BATH: 50 cfm, 4 ft, 2 elbows -> 4 in flex (Table M1505.4.4.2 allows 25 ft) · BATH whole-house fan: 50 cfm,
Dryer duct 4 in 90 elbow	5	5	EA	
Dryer duct 4 in rigid metal, smooth wall (M1502.4.2)	20	23	LF	from the electrician's dryer outlet in PRIMARY CLOSET: 11 ft · from the electrician's dryer outlet in MECH/STO
Dryer vent box (recessed)	2	2	EA	
Dryer wall cap with damper, no screen (M1502.3)	2	2	EA	
Exhaust duct 6 in insulated (80-110 cfm fans, whole-house fan: Table M1505.4.4.2)	4.4	5	LF	BATH whole-house fan: 70 cfm, 4 ft, 2 elbows -> 5 in flex (Table M1505.4.4.2 allows 15 ft)
Outdoor air inlet 4 sq in NFA, through-wall with screen (WA exhaust-only whole-house ventilation, one per habitable room)	7	7	EA	unit A: an outdoor-air inlet (4 sq in NFA) in each of 4 habitable spaces (WA exhaust-only systems) - window tr
Range hood duct 6-10 in rigid, sealed (M1503.3)	11	13	LF	6 in rigid for a 300 cfm hood at the electrician's range outlet (electrical/jobs/shaw/devices.json) (M1503.3);
Range hood wall / roof cap with damper	2	2	EA	6 in wall cap with damper
Wall / roof cap 6 in with backdraft damper	1	1	EA	BATH whole-house fan: wall cap with backdraft damper
Wall cap 4 in with backdraft damper	2	2	EA	PR BATH: wall cap with backdraft damper · BATH whole-house fan: wall cap with backdraft damper
Whole-house exhaust fan, continuous duty, <= 1.0 sone (M1505.4) - replaces one electrical allowance fan	2	2	EA	unit A: whole-house fan 64 cfm continuous (Qr 43 x Csystem 1.5, one fan = unbalanced, not distributed) in BATH
Whole-house ventilation control - 24 h timer / override, labelled (WA M1505.4.2)	2	2	EA	labelled "WHOLE HOUSE VENTILATION - LEAVE ON" 24 h control with manual override (WA M1505.4.2); the electricia

Start-up, tests and design

Commissioning

item	measured	order	unit	note
Equipment labels, WSEC certificate data, owner hand-over	2	2	EA	per dwelling unit
Load calculation + equipment selection for the permit (Manual J / S or the WSU sizing form)	1	1	EA	room-by-room loads for the permit (this takeoff is the draft)
Start-up, charge verification, controls set-up (per outdoor unit)	2	2	EA	30k system · 24k system
Ventilation flow test - whole-house + local exhaust (WA M1505.4.3 / M1505.4.4)	2	2	EA	unit A: whole-house 64 cfm + every local exhaust measured at the grille (WA M1505.4.3 / M1505.4.4) · unit B: w

Refrigerant piping

item	measured	order	unit	note
Nitrogen pressure test + evacuation to 500 micron (per system)	2	2	EA	per refrigerant system
Refrigerant - additional charge beyond the pre-charge (lb)	1.8	1.8	lb	17 oz past the pre-charged length [U: the maker's oz / ft] · 10 oz past the pre-charged length [U: the maker's

The checker

PRACTICE BED 2: a 6k head turns down to ~1,393 Btuh at 26 F against a 1,619 Btuh zone - over Manual S's 0.80 x load: it will cycle; carry this room from the zone next door or serve it with a slim ducted unit

PRACTICE BED 1: a 6k head turns down to ~1,393 Btuh at 26 F against a 867 Btuh zone - over Manual S's 0.80 x load: it will cycle; carry this room from the zone next door or serve it with a slim ducted unit

PRACTICE unit A: the 30k outdoor unit carries 30,117 Btuh at 26 F against a 12,106 Btuh load - 2.5x

PRACTICE unit B: the 24k outdoor unit carries 24,094 Btuh at 26 F against a 12,613 Btuh load - 1.9x

DATA design temperature: 26 F is a stand-in (STAND-IN: MJ8 Table 1A Friday Harbor AP 30 F / 75 F [V] shifted by Bellingham's RC-1 gap (-4 / +2 F) - pull Table RC-1 (Shaw is 3 mi from the Friday Harbor station)) - pull WSEC Table RC-1 for shaw island before this goes out

PRACTICE electrical: the 30k multi outdoor unit typically needs a 25 A circuit [U: the nameplate MOCP governs]; the electrical job carries 20 A

PRACTICE electrical: the 24k multi outdoor unit typically needs a 25 A circuit [U: the nameplate MOCP governs]; the electrical job carries 20 A

DATA electrical: the electrical job labels 16 device(s) with the other unit's name - e.g. its 'Range (unit B)' marked unit B sits in unit A's footprint (DINING); this takeoff follows the plan (job.json units) - tell the electrician before the panel schedule goes out

note: unit A: 2 bath / powder room(s) have no head - they are carried by the zone they open into; WSEC allows resistance heat up to 0.5 W/sf or 500 W if a bath needs its own (an option line, not in the base)

note: unit B: 1 bath / powder room(s) have no head - they are carried by the zone they open into; WSEC allows resistance heat up to 0.5 W/sf or 500 W if a bath needs its own (an option line, not in the base)

note: bath fans: 3 read from the electrical skill (electrical/jobs/shaw/devices.json) - the electrician supplies and wires them; HVAC ducts and caps each one

note: refrigerant: a split system installed from 1/1/2026 is an A2L (R-454B / R-32) unit under the EPA Technology Transitions rule; R-410A equipment is out. A2L units carry leak detection / mitigation per their listing - it rides with the equipment line

note: ductless: each head's programmable wireless controller meets WSEC R403.1.1; a wired wall controller per zone is an option line

note: electrical: one of its bath-fan allowances becomes the HVAC whole-house fan (supplied here) - strike it from the electrical allowance, keep the wiring and add the 24 h control

Labour and inspections

crew of 2: set 24.5 h / 4 d, rough 31.2 h / 5 d, startup 10.0 h / 2 d - 11 days at 6.25 h. crew minutes are PLACEHOLDERS (no task-level install hours were found: references/research-hvac-market.md 3.3) - set them from a sub or the crew

Mechanical rough-in (cover) - after framing, before insulation; line sets pressure-tested and in place; ducts, boots, exhaust ducts and caps; firestops at the garage separation

Ventilation flow test - before final; whole-house rate and every local exhaust measured (WA M1505.4)

Final mechanical - equipment set, started, labelled; equipment listing / HSPF2 on the certificate, controls, condensate, clearances

permit: San Juan County CD&P residential mechanical fee schedule - NOT FOUND in the 9/25 research (references/research-hvac-code.md 8): price it from the county sheet